DOMINATED TWINS IN TYPE A: A COMPLETE DECISION PROCEDURE, AND THE CENSUS FOR $n \leq 16$

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Research-repo commit `8dbeb45` (`proofs/2026-11-08-q68-typeA-closure.md`, scripts `proofs/prove284/`). Registry `proofs/registry/seedQ2-kl-cone.json`. Supersedes, as a reference, the notes of 11-07 and 11-08.

Summary. Let Γ be a left cell of a finite Coxeter group, and let M_Γ be its cell module at $v = 1$. A *dominated twin* is a pair $x \neq z$ in Γ with $L(z) \subseteq L(x)$ and $\tilde{\mu}(y, z) \geq \tilde{\mu}(y, x)$ for every $y \in \Gamma$ with $L(y) \not\subseteq L(x)$. A twin forces the Kazhdan–Lusztig cone of M_Γ to be non-maximal among invariant pointed cones.

This note collects the type-A story into one stable reference.

- Theorem 7, proved for all finite W: the twin condition only has to be checked on an explicit witness set. Parity is a consequence of it, not an extra hypothesis.
- Proposition 11: twin-ness depends only on the pair of P-tableaux. This is now proved from Knuth's theorem and the star-invariance of μ.
- The census for $n \leq 16$, Theorem 13: 0, 4, 26, 114 twin pairs for $n \leq 13$ and $n = 14, 15, 16$. Every verdict is a replayed certificate, except for *one* KL value in S_{16}. That value was computed by two independent implementations, and only two of the 462 verdicts use it.

Grades: Theorem 7 proved; Proposition 11 proved (modulo textbook RSK/KL facts); Proposition 12 proved; census computed (certificate-complete).

1. Setting

W is a finite Coxeter group with simple reflections S; $L(w) = \{s : sw < w\}$ and $R(w) = \{s : ws < w\}$. $P_{y,w}$ are the KL polynomials. For $y < w$, $\mu(y, w)$ is the coefficient of $q^{(\ell(w)-\ell(y)-1)/2}$ in $P_{y,w}$. The symmetric version is $\tilde{\mu}(a, b) = \mu(a, b)$ if $a < b$, $\mu(b, a)$ if $b < a$, and 0 otherwise.

The preorder $\leq_L$ is the transitive closure of "$\tilde{\mu}(y, w) \neq 0$ and $L(y) \not\subseteq L(w)$"; left cells are its classes. The cell module, in the convention of Goertzen–Williamson [GW, Ex. 4.7], has basis $(e_x)_{x \in \Gamma}$ and

$$(1 + s)e_x = 0 \ \ (s \in L(x)), \qquad (1 + s)e_x = 2e_x + \textstyle\sum_{y \in \Gamma,\, s \in L(y)} \tilde{\mu}(x, y)e_y \ \ (s \notin L(x)).$$

Definition 1. A *dominated twin* is a pair $x \neq z \in \Gamma$ with $L(z) \subseteq L(x)$ and

$$\text{(TW)} \qquad \tilde{\mu}(y, z) \geq \tilde{\mu}(y, x) \quad \text{for all } y \in \Gamma \text{ with } L(y) \not\subseteq L(x).$$

For simply-laced W, put $\operatorname{int} A = \{s \in A : \text{all neighbours of } s \text{ lie in } A\}$. The *witness sets* of x are

$$W_{\mathrm{i}}(x) = \{sx : s \in S,\ sx \in \Gamma,\ L(sx) \not\subseteq L(x)\}, \qquad W_{\mathrm{ii}}(x) = \{y \in \Gamma : L(y) \supsetneq L(x),\ y < x\}.$$

Type A. We write $w \in S_n$ in one-line notation, so that $s_i w$ swaps the *values* $i, i + 1$. Let $(P, Q) = \mathrm{RSK}(w)$. Then $L(w) = \mathrm{Des}P(w)$ and $R(w) = \mathrm{Des}Q(w)$, and the left cells are $\Gamma_Q = \{w : Q(w) = Q\}$. A *candidate pair* is a pair (x, z) in one cell with $L(x) = A$ and $\varnothing \neq L(z) \subseteq \operatorname{int} A$.

Inputs used without proof (textbook; all were validated in S_5/S_6 in earlier sessions):

- (K1) The KL recursion [KL, (2.2.c)], and $P_{y,w} = P_{sy,w}$ for $s \in L(w)$ [KL, (2.3.g)].
- (K2) $\mu \geq 0$ [KL80; Elias–Williamson].

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- (K3) $P_{y,w} = P_{y^{-1},w^{-1}}$.
- (K4) Diagram automorphisms preserve every $P_{y,w}$.
- (K5) $P_{y,w} = 1$ if $y < w$ with gap 1. Also $\tilde{\mu} = 0$ for an even gap, and for incomparable elements.
- (R) RSK and Schützenberger symmetry; left cells = $\{Q$ fixed$\}$; Knuth's theorem: $P(w) = P(w')$ iff w and w' are connected by elementary Knuth moves $yxz \leftrightarrow yzx$, $xzy \leftrightarrow zxy$ $(x < y < z)$.

2. The lemmas

Lemma 2 (F1, [KL, 2.3.e])**.** *Let $y < w$ and $s \in L(w) \setminus L(y)$. If $\mu(y,w) \neq 0$, then $y = sw$.*

Proof. By (K1), $P_{y,w} = P_{sy,w}$. If $sy \neq w$, then $sy < w$, so $\deg P_{sy,w} \leq (\ell(w) - \ell(y) - 2)/2$ and $\mu(y,w) = 0$. □

Lemma 3 (A)**.** *If (x,z) is a dominated twin, then $K = \mathrm{cone}(\{e_w\}_{w \neq z} \cup \{e_z - e_x\})$ is simplicial and $\{1+s\}$-invariant, and it strictly contains $C_{KL} = \mathrm{cone}\{e_w\}$.*

Proof. Put $v = e_z - e_x$. Then $e_z = v + e_x$, so $K \supseteq C_{KL}$; the containment is strict because $v \notin C_{KL}$. It remains to compute $(1+s)v$:

- if $s \in L(z) \subseteq L(x)$, it is 0;
- if $s \in L(x) \setminus L(z)$, it is $(1+s)e_z \in C_{KL}$;
- if $s \notin L(x)$, it is $2v + \sum_{s \in L(y)} (\tilde{\mu}(z,y) - \tilde{\mu}(x,y)) e_y$. Every such y has $L(y) \not\subseteq L(x)$ and $y \notin \{x,z\}$, so each coefficient is ≥ 0 by (TW). □

Lemma 4 (C1, C2)**.** *Let (x,z) be a twin, $s \in L(x)$, $t \notin L(x)$, and $m(s,t) = 3$. Then $s \notin L(z)$. Hence, if W is simply laced, $\varnothing \neq L(z) \subseteq \mathrm{int}\, L(x)$.*

Proof. Let ${}^*x \in \{sx, tx\}$ be the element of type $\{t\}$, that is, with $L({}^*x) \cap \{s,t\} = \{t\}$. Its left descent set is incomparable with $L(x)$ and $\tilde{\mu}({}^*x, x) = 1$, so ${}^*x \in \Gamma$, and (TW) gives $\tilde{\mu}({}^*x, z) \geq 1$.

Suppose $s \in L(z)$. Then z has type $\{s\}$. Apply Lemma 2 in whichever Bruhat direction holds: it gives $z \in \{s\,{}^*x,\, t\,{}^*x\}$. The element of type $\{s\}$ in this set is x, so $z = x$, a contradiction.

Finally, $L(z) \neq \varnothing$, since otherwise $z = e$, and $\{e\}$ is a cell of size one. (The pairing step is Stembridge's [St, Props. 4.2, 4.4].) □

Lemma 5 (S, star invariance; [KL, Thm. 4.2], proved in-file in PROVE 280)**.** *Let $m(s,t) = 3$ and $y, w \in D_L(s,t)$. Then $\tilde{\mu}(y,w) = \tilde{\mu}({}^*y, {}^*w)$. By (K3), the same holds for right stars on $D_R(s,t)$.*

Outline. Write elements of $D_L(s,t)$ as su, tsu, tu, stu, with u minimal in its coset.

- *Same coset:* check the six pairs directly.
- *Different types:* Lemma 2 forces $w = {}^*y$ on both sides.
- *Same form,* $(su, su') \mapsto (tsu, tsu')$: apply (K1) at tw with $w = su'$. The $qP_{tsu,su'}$ term and the single surviving sum term $z = stsu$ cancel.
- *Mixed form,* $(su, stu') \mapsto (tsu, tu')$, the gap-changing case: apply (K1) at $s \cdot tu'$. The first term has too low a degree, the second contributes $\mu(tsu, tu')$, and the sum is killed by Lemma 2.

The remaining cases follow by $s \leftrightarrow t$ symmetry and by reading identities right to left. □

Lemma 6 (Layer Lemma; folklore [BB, Thm. 6], [Po, L. 2.6])**.** *Let u be minimal in $W_K u$ and $v, v' \in W_K$. Then $P_{v'u,vu} = P_{v',v}$.*

Outline. Induct on $\ell(v)$ using (K1). Since $w \mapsto w^K$ is order-preserving [BB2, Prop. 2.5.1], every term of the recursion stays in the layer $W_K u$. □

3. Completeness of the witness set

Theorem 7. *Let W be finite, and let $x \neq z \in \Gamma$ with $L(z) \subseteq L(x) =: A$.*

(a) If $y \in \Gamma$, $L(y) \not\subseteq A$ and $\tilde{\mu}(y,x) \neq 0$, then $y \in W_{\mathrm{i}}(x) \cup W_{\mathrm{ii}}(x)$.
(b) $\tilde{\mu}(y,x) = 1$ for every $y \in W_{\mathrm{i}}(x)$.
(c) (x,z) is a dominated twin $\iff$ $\tilde{\mu}(y,z) \geq \tilde{\mu}(y,x)$ for all $y \in W_{\mathrm{i}}(x) \cup W_{\mathrm{ii}}(x)$.

Proof. (a) The elements y and x are comparable, and $y \neq x$.

- If $y > x$, take $r \in L(y) \setminus A$. Lemma 2 gives $y = rx$.
- If $y < x$ and $A \not\subseteq L(y)$, take $r \in A \setminus L(y)$. Lemma 2 gives $y = rx$.
- Otherwise $y < x$ and $L(y) \supsetneq A$, so $y \in W_{\mathrm{ii}}(x)$.

(b) holds by (K5).

(c) For $\Leftarrow$: every y with $L(y) \not\subseteq A$ that lies outside the witness sets has $\tilde{\mu}(y,x) = 0 \leq \tilde{\mu}(y,z)$, by (a) and (K2). The direction $\Rightarrow$ is trivial. □

Corollary 8 (parity is implied)**.** *Let W be connected and simply laced, and let (x,z) be a twin. Then $W_{\mathrm{i}}(x) \neq \varnothing$ and $\ell(x) \equiv \ell(z) \pmod 2$.*

Proof. By Lemma 4, $L(x) \neq \varnothing$. Also $L(x) \neq S$, because $\{w_0\}$ is a cell. Choose adjacent $s \in L(x)$ and $t \notin L(x)$. Then ${}^*x \in W_{\mathrm{i}}(x)$, and Theorem 7 gives $\tilde{\mu}({}^*x, z) \geq 1$. So $\ell({}^*x) - \ell(z)$ is odd, and $\ell({}^*x) = \ell(x) \pm 1$. □

Remark 9. Theorem 7 is close to a tautology: its content is Lemma 2. The real content of a twin verdict lies in the μ-values on the witness set. In particular, the decision procedure (the $C2$ filter, then the witness inequality) is complete. Parity is just its cheapest special case, namely an even gap at ${}^*x$.

Nothing excludes $\tilde{\mu}(y,x) \geq 2$ for $y \in W_{\mathrm{ii}}(x)$ a priori. In the census every such value that was needed was certified to be exactly 0 or 1.

4. Cell independence

Lemma 10. *In S_n, the right star ρ_j for $\{s_j, s_{j+1}\}$ is defined on w iff exactly one of $w(j) > w(j{+}1)$ and $w(j+1) > w(j+2)$ holds. When it is defined, it is the elementary Knuth move on positions $j, j+1, j+2$, and every Knuth move arises this way.*

Proof. Let $x < y < z$.

- For the pattern yxz: right multiplication by s_j gives xyz, which has type $\varnothing$, and by s_{j+1} gives yzx, which has type $\{s_{j+1}\}$. So $\rho_j(yxz) = yzx$.
- For the pattern zxy: s_j gives xzy, of type $\{s_{j+1}\}$, and s_{j+1} gives zyx, of type $\{s_j, s_{j+1}\}$. So $\rho_j(zxy) = xzy$.

The two type-$\{s_{j+1}\}$ patterns are the inverses of these moves. □

Proposition 11. *Let $Q, Q' \in \mathrm{SYT}(\lambda)$. There is a bijection $\varphi : \Gamma_Q \to \Gamma_{Q'}$ with $P(\varphi w) = P(w)$ and $\tilde{\mu}(\varphi w, \varphi w') = \tilde{\mu}(w, w')$. Hence $((T,Q),(T',Q))$ is a twin iff $((T,Q'),(T',Q'))$ is.*

Proof. By Knuth's theorem and Lemma 10, some sequence of right stars carries (P_0, Q) to (P_0, Q'). Since $R = \mathrm{Des}Q$ is constant on a cell, each $\rho = \rho_j$ in the sequence is defined on the whole current cell Γ. Fix one such step.

- ρ is injective on Γ.
- ρ preserves P, hence L.
- ρ preserves $\tilde{\mu}$, by Lemma 5.

For $w, w' \in \Gamma$, any chain realising $w \leq_L w'$ stays inside Γ. Applying ρ link by link gives $\rho w \leq_L \rho w'$, and symmetrically. So $\rho(\Gamma)$ lies in a single cell $\Gamma_{Q''}$. Its P-values exhaust $\mathrm{SYT}(\lambda)$, so $\rho(\Gamma) = \Gamma_{Q''}$. Composing the steps gives φ. The condition (TW) is expressed through L and $\tilde{\mu}$ alone, so it transports along φ. □

As a by-product, $Q(\rho_j w)$ depends only on $Q(w)$; this is Haiman's dual equivalence. The earlier connectivity computations (`dualconn.py`, random-Q re-decisions) become controls.

5. Certificates and the one computed value

Each census verdict has a certificate, replayed by code that shares nothing with the search code. A certificate is one of the following.

- *Star path plus endpoint.* A sequence of simultaneous left or right stars, with membership in D_L or D_R asserted at every step, so $\tilde{\mu}$ is constant along it by Lemma 5. At the endpoint, a comparable pair with gap 1 gives $\tilde{\mu} = 1$; an even gap, Bruhat-incomparability, or a violation of Lemma 2 gives $\tilde{\mu} = 0$.
- *Layer certificate* (5 pairs of shape $(5, 5, 3, 3)$). Apply the Layer Lemma to delete 16 (or 1), then finish with a star path in S_{15}.
- *Ext certificate* (2 pairs of shape $(6, 4, 3, 3)$). A right-star path to a pair whose μ was computed.

The verdict is TWIN when every witness has certified $\tilde{\mu}(y, z) = 1 = \tilde{\mu}(y, x)$ and no $\tilde{\mu}(y, x)$ with $y \in W_{\mathrm{ii}}$ is undecided. It is NO when some witness has $\tilde{\mu}(y, z) = 0 < \tilde{\mu}(y, x)$. By Theorem 7, both rules are exactly (TW).

Proposition 12. *Let*

$$x_0 = [5, 4, 3, 1, 9, 8, 7, 2, 15, 14, 11, 6, 13, 10, 12, 16],$$
$$z_0 = [14, 8, 4, 1, 15, 9, 5, 2, 16, 10, 6, 3, 11, 7, 12, 13],$$

with $\ell = 33, 52$, and let $c(w) = w_0 w w_0$. The two computed pairs used by the ext certificates are $\rho_3(x_0, z_0)$ (with $\ell = 34, 53$) and $\rho_{13}\rho_{11}(cx_0, cz_0)$ (with $\ell = 35, 54$). Hence both have $\mu = \mu(x_0, z_0)$.

Proof. The first move is the Knuth move $3, 1, 9 \mapsto 3, 9, 1$ in x_0 and $4, 1, 15 \mapsto 4, 15, 1$ in z_0. The second path was found by search and compared literally (`prove284/link.py`). Then apply Lemma 5 and (K4). □

The value $\mu(x_0, z_0) = 1$ is computed. Two independent implementations give the same result: Dyer's path formula (`dyer2.c`) and a from-scratch Deodhar parabolic KL on a 22,208-element W^J interval. Both give

$$P_{x_0, z_0} = 1 + 12q + 62q^2 + 181q^3 + 328q^4 + 381q^5 + 279q^6 + 119q^7 + 26q^8 + q^9.$$

The full Bruhat interval has 894,344 elements.

6. The census

Theorem 13 (computed, certificate-complete)**.** *Up to the choice of cell (Proposition 11), the dominated twins in S_n are as follows.*

- *$n \leq 13$: none, since there are no candidate pairs [P279, Thm. C3; exhaustive Kostka-dominance check].*
- *$n = 14$: 4 of the 4 candidate pairs.*
- *$n = 15$: 26 of 48. Of the 22 NO verdicts, 20 are parity kills and 2 are W_{ii}-kills.*
- *$n = 16$: 114 of 410. The 296 NO verdicts are 186 parity kills, 19 with a W_{ii}-kill, 86 with a W_{i}-kill only, and 5 layer certificates.*

The theorem holds modulo (K1)–(K5) and (R), the exhaustive enumeration code, the certificate replays, and the single value $\mu(x_0, z_0) = 1$. *That value is used only by the TWIN verdicts of the two* $(6, 4, 3, 3)$ *pairs* 353 *and* 364. *If it were* 0, *the count would be* 112/298.

$\lambda \vdash 15$	pairs	TW	NO	$\lambda \vdash 16$	pairs	TW	NO
6333	4	4	0	7333	10	10	0
5442	3	1	2	6442, 64411	6, 6	1, 1	5, 5
54411	3	1	2	6433	54	11[†]	43
5433	12	4	8	63331	12	4	8
53331	3	1	2	5542, 55411	3, 3	0, 0	3, 3
4443	4	4	0	5533	18	4	14[‡]
44421	12	6	6	5443	43	9	34
444111	4	4	0	54421, 544111	48, 12	6, 4	42, 8
44331	3	1	2	54331	48	6	42
				53332, 533311	3, 6	0, 1	3, 5
				4444	4	4	0
				44431	43	15	28
				44422, 444211	18, 54	9, 18	9, 36
				4441111	10	10	0
				44332, 443311	3, 6	0, 1	3, 5
total	48	26	22	total	410	114	296

[†]Two of these rest on $\mu(x_0, z_0)$. [‡]Five of these are by layer certificates. At $n = 14$ the four twin shapes are 4433, 5333, 4442 and 44411, each with a single candidate pair.

Replay (2026-11-08). Every certificate file for $n = 14, 15, 16$ was re-run today, and the counts above were reproduced (`prove284/out_rerun.txt`).

7. What died

claim	killed by	grade
simply laced $\Rightarrow$ no twins	$(4, 4, 3, 3)$ twin in S_{14}	certificate
$(\star)$-violation $\iff$ twin, at the (λ, A) level	22 NO pairs in S_{15}	certificate
violation $\iff$ twin, at the shape level	$53332, 44332, 5542, 55411$ in S_{16}	certificate
twin $\iff$ parity and $W_{\mathrm{ii}} = \varnothing$	86 false positives in S_{16}	certificate
(ii)-elements always kill	pairs $352, 353, 364, 365$	uses $\mu(x_0, z_0)$
Lemma I: $\mu(x, z) = 0$ for candidate pairs	$\mu = 1$ at five S_{16} instances	computed

One Lemma I counterexample does not depend on x_0:

$$x = [7, 3, 2, 11, 10, 6, 5, 1, 15, 9, 8, 4, 16, 14, 13, 12],$$
$$z = [10, 6, 2, 14, 11, 7, 3, 1, 15, 12, 8, 4, 16, 13, 9, 5],$$

with $P_{x,z} = 1 + 8q + 24q^2 + 35q^3 + 26q^4 + 9q^5 + q^6$. This was computed by two implementations, over an interval of 20,160 elements.

8. Gaps

(1) $\mu(x_0, z_0) = 1$ is computed, not proved.
(2) (K1)–(K5) and (R) are used as textbook facts.
(3) The enumeration codes are trusted to be exhaustive. They are cross-checked at small n.
(4) By decision, this note goes no further than $n = 16$ and proves no structural twin criterion beyond Theorem 7.

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